

5.2 Linear Equations in One Variable with Applications



Figure 5.4 Most gyms have a monthly membership fee. (credit: modification of work "Morning PT after the Holidays 2021" by Fort Drum & 10th Mountain Division (LI)/Flickr, Public Domain Mark 1.0)

Learning Objectives

After completing this section, you should be able to:

1. Solve linear equations in one variable using properties of equations.
2. Construct a linear equation to solve applications.
3. Determine equations with no solution or infinitely many solutions.
4. Solve a formula for a given variable.

In this section, we will study linear equations in one variable. There are several real-world scenarios that can be represented by linear equations: taxi rentals with a flat fee and a rate per mile; cell phone bills that charge a monthly fee plus a separate rate per text; gym memberships with a monthly fee plus a rate per class taken; etc. For example, if you join your local gym at \$10 per month and pay \$5 per class, how many classes can you take if your gym budget is \$75 per month?

Linear Equations and Applications

Solving any equation is like discovering the answer to a puzzle. The purpose of solving an equation is to find the value or values of the variable that makes the equation a true statement. Any value of the variable that makes the equation true is called a solution to the equation. It is the answer to the puzzle! There are many types of equations that we will learn to solve. In this section, we will focus on a **linear equation**, which is an equation in one variable that can be written as

$$ax + b = 0$$

where a and b are real numbers and $a \neq 0$, such that a is the coefficient of x and b is the constant.

To solve a linear equation, it is a good idea to have an overall strategy that can be used to solve any linear equation. In the Example 5.12, we will give the steps of a general strategy for solving any linear equation. Simplifying each side of the equation as much as possible first makes the rest of the steps easier.

EXAMPLE 5.12

Solving a Linear Equation Using a General Strategy

Solve $7(n - 3) - 8 = -15$

 Solution

Step 1: Simplify each side of the equation as much as possible.	Use the Distributive Property. Notice that each side of the equation is now simplified as much as possible.	$7(n - 3) - 8 = -15$ $7n - 21 - 8 = -15$ $7n - 29 = -15$
Step 2: Collect all variable terms on one side of the equation.	Nothing to do; all n -terms are on the left side.	$7n - 29 = -15$
Step 3: Collect constant terms on the other side of the equation.	To get constants only on the right, add 29 to each side. Simplify.	$7n - 29 + 29 = -15 + 29$ $7n = 14$
Step 4: Make the coefficient of the variable term equal to 1.	Divide each side by 7. Simplify.	$\frac{7n}{7} = \frac{14}{7}$ $n = 2$
Step 5: Check the solution.	Let $n = 2$ Subtract.	Check: $7(n - 3) - 8 = -15$ $7(2 - 3) - 8 \stackrel{?}{=} -15$ $7(-1) - 8 \stackrel{?}{=} -15$ $-7 - 8 \stackrel{?}{=} -15$ $-15 = -15 \checkmark$

 YOUR TURN 5.12

1. Solve $2(x + 1) - 3 = 5$

In [Example 5.12](#), we used both the addition and division property of equations. All the properties of equations are summarized in table below. Basically, what you do to one side of the equation, you must do to the other side of the equation to preserve equality.

Operation	Property	Example
Addition	If $a = b$ Then $a + c = b + c$	$2 = 2$ $2 + 3 = 2 + 3$ $5 = 5$
Subtraction	If $a = b$ Then $a - c = b - c$	$5 = 5$ $5 - 2 = 5 - 2$ $3 = 3$

Operation	Property	Example
Multiplication	If $a = b$ Then $a \bullet c = b \bullet c$	$3 = 3$ $3 \bullet 4 = 3 \bullet 4$ $12 = 12$
Division	If $a = b$ Then $a \div c = b \div c$ for $c \neq 0$	$8 = 8$ $8 \div 2 = 8 \div 2$ $4 = 4$

 Be careful to multiply and divide every term on each side of the equation. For example, $2 + x = \frac{x}{3}$ is solved by multiplying BOTH sides of the equation by 3 to get $3(2 + x) = 3\left(\frac{x}{3}\right)$ which gives $6 + 3x = x$. Using parentheses will help you remember to use the distributive property! A division example, such as $3(x + 2) = 6x + 9$, can be solved by dividing BOTH sides of the equation by 3 to get $\frac{3(x + 2)}{3} = \frac{6x + 9}{3}$, which then will lead to $x + 2 = 2x + 3$.

EXAMPLE 5.13

Solving a Linear Equation Using Properties of Equations

Solve $9(y - 2) - y = 16 + 7y$.

Solution

Step 1: Simplify each side.

$$9(y - 2) - y = 16 + 7y$$

$$9y - 18 - y = 16 + 7y$$

$$8y - 18 = 16 + 7y$$

Step 2: Collect all variables on one side.

$$8y - 18 - 7y = 16 + 7y - 7y$$

$$y - 18 = 16$$

Step 3: Collect constant terms on one side.

$$y - 18 + 18 = 16 + 18$$

$$y = 34$$

Step 4: Make the coefficient of the variable 1. Already done!

Step 5: Check.

$$9(34) - 18 - (34) \stackrel{?}{=} 16 + 7(34)$$

$$306 - 18 - 34 \stackrel{?}{=} 16 + 238$$

$$288 - 34 \stackrel{?}{=} 254$$

$$254 = 254 \checkmark$$

YOUR TURN 5.13

1. Solve $6(y - 2) - 5y = 4(y + 3) - 4(y - 1)$.

WHO KNEW?

Who Invented the Symbol for Equals ?

Before the creation of a symbol for equality, it was usually expressed with a word that meant equals, such as *aequales* (Latin), *esgale* (French), or *gleich* (German). Welsh mathematician and physician Robert Recorde is given credit for inventing the modern sign. It first appears in writing in *The Whetstone of Witte*, a book Recorde wrote about algebra, which was published in 1557. In this book, Recorde states, "I will set as I do often in work use, a pair of parallels, or Gemowe (twin) lines of one length, thus: $==$, because no two things can be more equal." Although his version of the sign was a bit longer than the one we use today, his idea stuck and "=" is used throughout the world to indicate equality in mathematics.

In [Algebraic Expressions](#), you translated an English sentence into an equation. In this section, we take that one step further and translate an English paragraph into an equation, and then we solve the equation. We can go back to the opening question in this section: *If you join your local gym at \$10 per month and pay \$5 per class, how many classes can you take if your gym budget is \$75 per month?* We can create an equation for this scenario and then solve the equation (see [Example 5.15](#)).

EXAMPLE 5.14

Constructing a Linear Equation to Solve an Application

The Beaudrie family has two cats, Basil and Max. Together, they weigh 23 pounds. Basil weighs 16 pounds. How much does Max weigh?

Solution

Let b = Basil's weight and m = Max's weight.

$$b + m = 23$$

We also know that Basil weighs 16 pounds so:

Steps 1 and 2: $16 + m = 23$

Since both sides are simplified, the variable is on one side of the equation, we start in Step 3 and collect the constants on one side:

Step 3:

$$\begin{aligned} 16 + m - 16 &= 23 - 16 \\ m &= 7 \end{aligned}$$

Step 4: is already done so we go to Step 5:

Step 5:

$$\begin{aligned} 16 + 7 &\stackrel{?}{=} 23 \\ 23 &= 23 \checkmark \end{aligned}$$

Basil weighs 16 pounds and Max weighs 7 pounds.

YOUR TURN 5.14

- Sam and Henry are roommates. Together, they have 68 books. Sam has 26 books. How many books does Henry have?

EXAMPLE 5.15**Constructing a Linear Equation to Solve Another Application**

If you join your local gym at \$10 per month and pay \$5 per class, how many classes can you take if your gym budget is \$75 per month?

 Solution

If we let x = number of classes, the expression $5x + 10$ would represent what you pay per month if each class is \$5 and there's a \$10 monthly fee per class. \$10 is your constant. If you want to know how many classes you can take if you have a \$75 monthly gym budget, set the equation equal to 75. Then solve the equation $5x + 10 = 75$ for x .

Steps 1 and 2:

$$5x + 10 = 75$$

Step 3:

$$\begin{aligned} 5x + 10 - 10 &= 75 - 10 \\ 5x &= 65 \end{aligned}$$

Step 4:

$$\begin{aligned} \frac{5x}{5} &= \frac{65}{5} \\ x &= 13 \end{aligned}$$

Step 5:

$$\begin{aligned} 5(13) + 10 &\stackrel{?}{=} 75 \\ 65 + 10 &\stackrel{?}{=} 75 \\ 75 &= 75 \checkmark \end{aligned}$$

The solution is 13 classes. You can take 13 classes on a \$75 monthly gym budget.

**YOUR TURN 5.15**

1. On June 7, 2021, the national average price for regular gasoline was \$3.053 per gallon. If Aiko fills up his car with 16 gallons, how much is the total cost? Round to the nearest cent.

EXAMPLE 5.16**Constructing an Application from a Linear Equation**

Write an application that can be solved using the equation $50x + 35 = 185$. Then solve your application.

 Solution

Answers will vary. Let's say you want to rent a snowblower for a huge winter storm coming up. If x = the number of days you rent a snowblower, then the expression $50x + 35$ represents what you pay if, for each day, it costs \$50 to rent the snowblower and there is a \$35 flat rental fee. \$35 is the constant. To find out how many days you can rent a snowblower for \$185, set the expression equal to 185. Then solve the equation $50x + 35 = 185$ for x .

Steps 1 and 2:

$$50x + 35 = 185$$

Step 3:

$$50x + 35 - 35 = 185 - 35$$

$$50x = 150$$

Step 4:

$$\frac{50x}{50} = \frac{150}{50}$$

$$x = 3$$

Step 5:

$$50(3) + 35 \stackrel{?}{=} 185$$

$$150 + 35 \stackrel{?}{=} 185$$

$$185 = 185 \checkmark$$

The equation is $50x + 35 = 185$ and the solution is 3 days. You can rent a snowblower for 3 days on a \$185 budget.

YOUR TURN 5.16

1. Write an application that can be solved using the equation $25x + 75 = 200$. Then solve your application.

Linear Equations with No Solutions or Infinitely Many Solutions

Every linear equation we have solved thus far has given us one numerical solution. Now we'll look at linear equations for which there are no solutions or infinitely many solutions.

EXAMPLE 5.17

Solving a Linear Equation with No Solution

Solve $3(x + 4) = 4x + 8 - x$.

Solution

Step 1: Simplify each side. $3(x + 4) = 4x + 8 - x$

$$3x + 12 - 3x = 3x + 8 - 3x$$

Step 2: Collect all variables to one side. $3x + 12 - 3x = 3x + 8 - 3x$

$$12 = 8$$

The variable x disappeared! When this happens, you need to examine what remains. In this particular case, we have $12 = 8$, which is not a true statement. When you have a false statement, then you know the equation has no solution; there does not exist a value for x that can be put into the equation that will make it true.

YOUR TURN 5.17

1. Solve $2(x + 6) = 3x + 4 - (x + 5)$.

EXAMPLE 5.18**Solving a Linear Equation with Infinitely Many Solutions**Solve $2(x + 5) = 4(x + 3) - 2x - 2$. **Solution****Step 1:**

$$2(x + 5) = 4(x + 3) - 2x - 2$$

$$2x + 10 = 4x + 12 - 2x - 2$$

$$2x + 10 = 2x + 10$$

Step 2:

$$2x + 10 - 2x = 2x + 10 - 2x$$

$$10 = 10$$

As with the previous example, the variable disappeared. In this case, however, we have a true statement ($10 = 10$). When this occurs we say there are infinitely many solutions; any value for x will make this statement true.

 YOUR TURN 5.18

1. Solve $3x - 7 - (x + 5) = 2(x - 6)$.

Solving a Formula for a Given Variable

You are probably familiar with some geometry formulas. A **formula** is a mathematical description of the relationship between variables. Formulas are also used in the sciences, such as chemistry, physics, and biology. In medicine they are used for calculations for dispensing medicine or determining body mass index. Spreadsheet programs rely on formulas to make calculations. It is important to be able to manipulate formulas and solve for specific variables.

To solve a formula for a specific variable means to isolate that variable on one side of the equal sign with a coefficient of 1. All other variables and constants are on the other side of the equal sign. To see how to solve a formula for a specific variable, we will start with the distance, rate, and time formula.

EXAMPLE 5.19**Solving for a Given Variable with Distance, Rate, and Time**Solve the formula $d = rt$ for t . This is the distance formula where d = distance, r = rate, and t = time. **Solution**Divide both sides by r : $d/r = rt/r$

$$d/r = t$$

 YOUR TURN 5.19

1. Solve the formula $I = Prt$ for t . This formula is used to calculate simple interest I , for a principal P , invested at a rate r , for t years.

 VIDEO

Solving for a Variable in an Equation (https://openstax.org/r/Solving_for_a_variable)

EXAMPLE 5.20**Solving for a Given Variable in the Area Formula for a Triangle**

Solve the formula $A = \frac{1}{2}bh$ for h . This is the area formula of a triangle where A = area, b = base, and h = height.

✓ **Solution**

Step 1: Multiply both sides by 2.

$$2A = 2(\frac{1}{2}bh)$$

$$2A = bh$$

Step 2: Divide both sides by b .

$$\frac{2A}{b} = \frac{bh}{b}$$

$$\frac{2A}{b} = h$$

$$h = \frac{2A}{b}$$

> YOUR TURN 5.20

1. Solve the formula $V = \frac{1}{3}\pi r^2 h$ for h . This formula is used to calculate the volume V of a right circular cone with radius r and height h .

WORK IT OUT**Using Algebra to Understand Card Tricks**

You will need to perform this card trick with another person. Before you begin, the two people must first decide which of the two will be the *Dealer* and which will be the *Partner*, as each will do something different. Once you have decided upon that, follow the steps here:

Step 1: Dealer and Partner: Take a regular deck of 52 cards, and remove the face cards and the 10s.

Step 2: Dealer and Partner: Shuffle the remaining cards

Step 3: Dealer and Partner: Select one card each, but keep them face down and don't look at them yet.

Step 4: Dealer: Look at your card (just the Dealer!). Multiply its value by 2 (Aces = 1).

Step 5: Dealer: Add 2 to this result.

Step 6: Dealer: Multiply your answer by 5.

Step 7: Partner: Look at your card.

Step 8: Partner: Calculate: 10 - your card, and tell this information to the dealer.

Step 9: Dealer: Subtract the value the Partner tells you from your total to get a final answer.

Step 10: Dealer: verbally state the final answer.

Step 11: Dealer and Partner: Turn over your cards. Now, answer the following questions

1. Did the trick work? How do you know?
2. Why did this occur? In other words, how does this trick work?

Check Your Understanding

9. Is the solution strategy used in solving the linear equation correct? If it is correct, show the final step (check the solution). If it is not correct, explain why.

$$\begin{aligned}
 8(x - 2) &= 6(x + 10) \\
 8x - 16 &= 6x + 60 \\
 8x - 16 - 6x &= 6x + 60 - 6x \\
 2x - 16 + \mathbf{16} &= 60 + \mathbf{16} \\
 2x &= 76 \\
 x &= 38
 \end{aligned}$$

10. Is the solution strategy used in solving the linear equation correct? If it is correct, show the final step (check the solution). If it is not correct, explain why.

$$\begin{aligned}
 7 + 4(2 + 5x) &= 3(6x + 7) - (13x + 36) \\
 7 + 8 + 20x &= 18x + 21 - 13x - 36 \\
 15 + 20x &= 5x - 15 \\
 15 + 20x - \mathbf{5x} &= 5x - 15 - \mathbf{5x} \\
 15 + 15x - \mathbf{15} &= -15 - \mathbf{15} \\
 15x &= -30 \\
 x &= -2
 \end{aligned}$$

11. Is the solution strategy used in solving the linear equation correct? If it is correct, show the final step (check the solution). If it is not correct, explain why.

$$\begin{aligned}
 8x + 7 - (2x - 9) &= 22 - (4x - 4) \\
 8x + 7 - 2x - 9 &= 22 - 4x - 4 \\
 6x - 2 &= 18 - 4x \\
 6x - 2 + \mathbf{4x} &= 18 - 4x + \mathbf{4x} \\
 10x - 2 + \mathbf{2} &= 18 + \mathbf{2} \\
 10x &= 20 \\
 x &= 2
 \end{aligned}$$

For the following exercises, use this scenario: The Nice Cab Company charges a flat rate of \$3.00 for each fare, plus \$1.70 per mile. A competing taxi service, the Enjoyable Cab Company, charges a flat rate of \$5.00 for each fare, plus \$1.60 per mile.

12. Using the variable x for number of miles, write the equation that would allow you to find the total fare (T) using the Nice Cab Company.
13. It is 22 miles from the airport to your hotel. What would be your total fare using the Nice Cab Company?
14. Using the variable y for number of miles, write the equation that would allow you to find the total fare (T) using the Enjoyable Cab Company.
15. Using the same 22-mile trip from the airport to the hotel, how much would the total fare be for using the Enjoyable Cab Company?
16. Based on the cost of each cab ride, which cab company should you use for the trip from the airport to the hotel? Why?
17. After solving the linear equation $3(2x - 3) = 12(x - 3) - 3(2x - 9)$, Nancy says there is no solution. Luis believes there are infinitely many solutions. Who is right?
18. The conversion formula between the Fahrenheit temperature scale and the Celsius temperature scale is given by this formula: $C = \frac{5}{9}(F - 32)$, where C is the temperature in degrees Celsius and F is the temperature in degrees Fahrenheit. What is the correct formula when solved for F ?
 - a. $F = \frac{5}{9}C - 32$
 - b. $F = \frac{9}{5}C - 32$
 - c. $F = \frac{5}{9}C + 32$
 - d. $F = \frac{9}{5}C + 32$
19. To find a temperature on the Kelvin temperature scale, add 273 degrees to the temperature in Celsius. Which formula illustrates this?
 - a. $C = K + 273$
 - b. $K = C + 273$
 - c. $K = C - 273$

- d. $C = K - 273$
20. Using the information from exercise 18 and exercise 19, which conversion formula would you use to find degrees Kelvin when given degrees Fahrenheit?
- $K = \frac{5}{9}(F - 32) + 273$
 - $K = \frac{5}{9}F + 241$
 - $K = \frac{9}{5}(F - 32) + 273$
 - $K = \frac{9}{5}F + 241$
21. There is a fourth temperature scale, although it is not used much today. The Rankin temperature scale varies from the Fahrenheit scale by about 460 degrees. So given a temperature in Fahrenheit, add 460 degrees to get the temperature in Rankin. Which formula represents a formula to find degrees Rankin when given degrees Celsius?
- $R = \frac{5}{9}C - 492$
 - $R = \frac{9}{5}C + 492$
 - $R = C + 492$
 - $R = \frac{5}{9}(C - 492)$



SECTION 5.2 EXERCISES

For the following exercises, solve the linear equations using a general strategy.

- $-(t - 19) = 28$
- $51 + 5(4 - q) = 56$
- $-6 + 6(5 - k) = 15$
- $3(10 - 2x) + 54 = 0$
- $-2(11 - 7x) + 54 = 4$

For the following exercises, solve the linear equations using properties of equations.

- $-12 + 8(x - 5) = -4 + 3(5x - 2)$
- $4(p - 4) - (p + 7) = 5(p - 3)$
- $3(a - 2) - (a + 6) = 4(a - 1)$
- $4[5 - 8(4c - 3)] = 12(1 - 13c) - 8$
- $5[9 - 2(6d - 1)] = 11(4 - 10d) - 139$

For the following exercises, construct a linear equation to solve an application.

- It costs \$0.55 to mail one first class letter. Construct a linear equation and solve to find how much it costs to mail 13 letters.
- Normal yearly snowfall at the local ski resort is 12 inches more than twice the amount it received last season. The normal yearly snowfall is 62 inches. Construct a linear equation and solve to find what the snowfall was last season.
- Guillermo bought textbooks and notebooks at the bookstore. The number of textbooks was three more than twice the number of notebooks. He bought seven textbooks. Construct a linear equation and solve to find how many notebooks he bought.
- Gerry worked Sudoku puzzles and crossword puzzles this week. The number of Sudoku puzzles he completed is eight more than twice the number of crossword puzzles. He completed 22 Sudoku puzzles. Construct a linear equation and solve to find how many crossword puzzles he did.
- Laurie has \$46,000 invested in stocks. The amount invested in stocks is \$8,000 less than three times the amount invested in bonds. Construct a linear equation and solve to find how much Laurie invested in bonds.

For the following exercises, construct an application from a linear equation.

- $1,000x + 2,500 = 16,500$.
- $0.36t$ for $t = 333$.
- $150n + 120 = 570$.
- $4c + 2c$ for $c = 5$.
- $2s + 10 = 24$.

For the following exercises, state whether each equation has exactly one solution, no solution, or infinitely many solutions.

- $23z + 19 = 3(5z - 9) + 8z + 46$